Conceptual Prototype for Agricultural Waste Minimization in Asia (Group Assignment)

Course: SSPG2113 Creativity and Innovation

Group Size: 6 students per group

Weightage: 50% of Final Course Mark

Exhibition Date (Prototype): 9th January 2025

Submission Deadline (Report): 16th January 2025

1. Assignment Objective

The objective of this assignment is for your group to develop a conceptual prototype and a long-term strategic plan for a novel solution aimed at **significantly reducing or eliminating an agricultural or aquaculture waste stream in Asia**.

This assignment emphasizes the application of **creative techniques (SCAMPER)**, the analysis of the solution's competitive position (**SWOT**), and the accurate categorization of the innovation (**Four Dimensions of Innovation**), all culminating in a phased strategic roadmap.

2. Assignment Task: Waste Innovation Project

Step 1: Project Allocation and Problem Definition

Your group need to choose any agricultural waste produced in Asia.

- You must clearly define the scope of the problem, the specific type of waste (e.g., rice straws or palm oil fronds or pineapple crowns), and its current negative environmental or economic impact within a defined Asian region.

Step 2: Conceptual Prototype Development & SCAMPER Analysis

Design a **Conceptual Prototype** for a product, service, or system to address your assigned waste problem.

- **SCAMPER Requirement:** You must demonstrate how the **SCAMPER technique** was used in the ideation phase. Show at least **four distinct SCAMPER actions** that were applied to an existing waste management method or a similar product to generate your final, novel prototype concept.

Step 3: Comprehensive Strategy Report

The core of your assignment is a written report detailing the strategic analysis of your conceptual prototype. (3000–3500 words, excluding appendices).

A. Four Dimensions of Innovation Categorization

Analyze your conceptual prototype using the four required categories of innovation. You must justify your selection for each pair:

Innovation Pair	Selection (Choose Three)	Justification (Why?)
1. Focus	Product OR Process	Is the core innovation in *what* is delivered (the product itself) or *how* it is made/delivered (the process)?
2. Magnitude	Incremental OR Radical	Does it build upon existing knowledge (Incremental) or require new knowledge and skills (Radical)?
3. Scope	Component OR Architectural	Does it change only one part of an existing system (Component) or change how all components interact (Architectural)?
4. Effect	Competence-Enhancing OR Competence-Destroying	Does it leverage the skills of existing firms (Enhancing) or render them obsolete (Destroying)?

B. SWOT Analysis and Prototype Advantages

Conduct a comprehensive **SWOT Analysis** (Strengths, Weaknesses, Opportunities, Threats) for your conceptual prototype and its intended market entry.

- **Benefits/Advantages Integration:** Within your analysis, you must explicitly detail the **benefits and advantages** of your prototype. These advantages should be linked to your **Strengths** (e.g., unique features, low operating cost) and **Opportunities** (e.g., environmental impact, job creation, economic value from waste).

C. Strategic Planning for Prototype Future (Roadmap)

Develop a phased strategic roadmap for the commercialization of your prototype, clearly defining goals, activities, and resource needs for three distinct phases:

1. **Short Term (0–18 months):** Focus on concept validation, small-scale testing, seeking initial funding, and finalizing design (e.g., "Build and test the V1 minimum viable product (MVP) in a controlled lab environment").
2. **Middle Term (18 months – 5 years):** Focus on scaling production, securing large contracts, formalizing partnerships, and achieving regional market penetration (e.g., "Establish a pilot facility in Johor and secure three major industrial partners").
3. **Long Term (5–10 years):** Focus on continuous innovation (next-gen features), achieving market leadership, and international expansion (e.g., "Licensing the technology across ASEAN countries").

Report Structure Requirement (Mandatory Sections):

The written report must include a dedicated final section: **Conclusion**. This conclusion must summarize the findings of your SWOT and Innovation Categorization, and re-state the long-term strategic viability of your conceptual prototype in the Asian context.

3. Deliverables

Deliverable	Weightage	Description
Written Report	**30%**	Formal, structured report (3000–3500 words) detailing the SCAMPER analysis, Innovation Categorization, SWOT/Advantages, Strategic Roadmap, and Conclusion.

Conceptual Prototype Poster	**20%**	A single **A1 size poster** (594 x 841 mm) that serves as the visual exhibit. It must summarize the key findings from the written report (SCAMPER logic, SWOT highlights, Innovation Type, and Strategic Roadmap). Must include a clear visual mock-up of the prototype.

4. Marking Rubric (50% Total Mark)

Criteria	Weight (%)	Description of Excellence
SCAMPER Technique Application	20%	Clear evidence of **four distinct SCAMPER actions** used to generate the final prototype concept. The analysis demonstrates how the technique led to a novel idea.
Innovation Categorization	15%	**All three dimensions are correctly identified and robustly justified.** Justifications are detailed, link directly to the course material, and demonstrate a deep strategic understanding of the innovation's nature.
SWOT Analysis & Prototype Advantages	20%	Comprehensive SWOT matrix. **Explicitly details the benefits/advantages of the prototype**, linking them logically to internal Strengths and external Opportunities.
Strategic Planning Roadmap	20%	**Roadmap is realistic, detailed, and clearly phased** (Short, Middle, Long Term). Goals, activities, and required resources are logical and align with the proposed scale-up strategy for the Asian market.
Conceptual Prototype & Poster Design	20%	Poster is well-designed (A1 size), visually compelling, and logically summarizes all key analysis points (SCAMPER, SWOT, Roadmap). The prototype visual is clear and effectively communicates core functionality.

Clarity, Professionalism, & Conclusion	5%	Report is well-written, follows academic structure, adheres to the word count, and uses professional referencing. **Conclusion effectively summarizes the strategic viability** of the project.
TOTAL	**100%**	